

ENTRY 025 Isolating the Compounding Formula — August 13, 2026 — Status: COMPLETE

Entry 025 — The Forgiveness Ratio Has a Closed Form

A Question Left Open Since Entry 016 Turns Out to Have an Exact Answer, Derived From First Principles and Verified Against Aer

1. What Was Done

Entry 024 confirmed that routing and real gate count are not the source of v3/v4's remaining error, leaving one candidate standing: the multiplicative compounding formula $(1 - p)^n$ itself, flagged as suspect in Entries 015 and 016 and never resolved. Testing it against FakeKyiv's full noise model conflates three effects at once — gate error, decoherence, and readout. This entry isolates one: an Aer noise model carrying **only** a depolarizing error on a single real Kyiv edge, no thermal relaxation, no readout error, applied n times to a cancelling-pair Bell circuit — the same method Entries 017–021 used, now aimed at a controlled synthetic channel instead of the full hardware model.

2. Result — a Confirmed Mismatch

n gates	Aer (isolated)	naive $(1-p)^n$	ratio	diff, pts
1	99.435%	99.014%	1.0043	+0.42
9	95.541%	91.468%	1.0445	+4.07
21	90.292%	81.213%	1.1118	+9.08
51	79.965%	60.329%	1.3255	+19.64
75	73.709%	47.560%	1.5498	+26.15

Edge (77,78), raw calibrated error 0.986%. Even with decoherence and readout completely removed, a chain of the identical depolarizing gate does not compound as $(1 - p)^n$. At 75 gates the gap is 26 points. Fitting an effective decay constant to the Aer curve gives 0.41% per gate against a nominal 0.99% — a ratio of **0.4156**, in the same range as the forgiveness ratios measured in Entries 017–021.

3. A Closed Form

Qiskit's two-qubit depolarizing_error(p) channel does exactly one of two things per application: with probability $(1 - p)$ it applies the ideal gate; with probability p it replaces the state with the maximally mixed state on 2 qubits. The maximally mixed state is an **absorbing state** — depolarizing it again changes nothing. So after n applications: the state never depolarized with probability $(1-p)^n$, contributing its full population to the {00,11} ideal set; it depolarized at least once with probability $1-(1-p)^n$, and a maximally mixed 2-qubit state distributes evenly across 4 basis outcomes, so exactly half of that population still lands in {00,11} by chance.

$$\text{success} = (1-p)^n \times 1 + [1-(1-p)^n] \times 0.5 = 0.5 + 0.5(1-p)^n$$

4. Verification

n	Aer measured	$0.5 + 0.5(1-p)^n$	diff, pts
5	97.519%	97.583%	-0.06
21	90.292%	90.607%	-0.31
51	79.965%	80.164%	-0.20
75	73.709%	73.780%	-0.07

Every point across nine gate counts (1 to 75) matches within 0.35 points — consistent with pure shot noise at 4 seeds $\times$ 8192 shots. Cross-checked on a second, unrelated edge (23,24), raw error 4.32% — 4.4 $\times$ the first edge's magnitude:

n	Aer measured	predicted	diff, pts
5	89.749%	90.088%	-0.34
21	69.791%	69.767%	+0.02
51	55.124%	55.250%	-0.13

Same closed form, same tolerance, an entirely different error magnitude. This is not curve-fitting — it is a derivation confirmed by measurement.

5. What This Means for QuantumBridge

The forgiveness ratios empirically fitted in Entries 017–021 were never purely a chip-physics phenomenon. For a Bell-state circuit, a large part of that ratio is a fixed geometric fact: **a 2-outcome-of-4 ideal set absorbs roughly half of any depolarizing error**, independent of calibration data, chip identity, or gate duration. It follows directly from measuring against a multi-outcome ideal set — exactly the mechanism flagged qualitatively in earlier entries, now given an exact formula.

This does not retire the empirical power law. The closed-form ratio approaches a floor of 0.5 as n grows; Entry 021's fitted law (coefficient 0.0898, exponent −0.3873) predicts ratios below 0.5 that keep falling with error magnitude, with no such floor. The gap between the two is where real chip physics — decoherence, correlated errors, whatever Entry 021's D-screen was isolating — actually lives. Entries 017–021 were fitting a curve that mixes a universal geometric constant with genuine hardware behavior, without separating them.

Entry 025 — Conclusion

The compounding-formula question left open by Entries 015 and 016 has an exact answer for the circuits this project has run throughout: $\text{success} = 0.5 + 0.5(1-p)^n$, not $(1-p)^n$. It follows from a two-line argument about how a depolarizing channel behaves once it has already fired once, and it matches Aer's own noise model to within shot noise across two edges spanning a 4.4× range in error magnitude. The naive multiplicative formula was never approximately right — it was measuring the wrong quantity from the start, decoherence and readout aside entirely. What remains genuinely open is how much of Entries 017–021's fitted power-law shape is this fixed geometric fact and how much is real hardware physics underneath it.

6. Next Step — Entry 026

Subtract the closed-form baseline from Entries 017–021's raw measurements and refit the residual. If a clean power law remains, that residual — not the mixed quantity fitted so far — is the real chip-physics signal, and it should replace `variable_forgiveness_ratio()` in v4. If nothing structured remains, the empirical fits from Entries 017–021 were mostly measuring this geometric artifact, and the decoherence term already added in Entries 023–024 is doing the real physical work.

Research Log — Index Addendum (Entry 025)

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Entry 025	Isolating the Compounding Formula — a Closed-Form Forgiveness Baseline	Aug 13, 2026	Complete

Entries 015 and 016's open question is resolved: the compounding formula was wrong, independent of decoherence and readout, and is now wrong by a known, derived amount. Entries 017-021's fitted power law is not withdrawn but is now understood to be a mixture of this geometric baseline and unseparated hardware physics.